

$$\textcircled{1} \quad f(x) = e^{3x^2+1}$$

$$y = 3x^2 + 1$$

$$f(x) = e^y$$

$$f'(x) = e^y \cdot (6x)$$

$$= \underline{6x \cdot e^{3x^2+1}}$$

$$\textcircled{2} \quad f(x) = 3x^4 + \frac{2}{x^2} - 5\sqrt{x}$$

$$f'(x) = 12x^3 - 4x^{-3} - \frac{5}{2}x^{-1/2}$$

$$\textcircled{3} \quad \sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\text{To show } \rightarrow \sigma'(x) = \sigma(x)(1 - \sigma(x))$$

$$\Rightarrow \sigma(x) = (1 + e^{-x})^{-1}$$

$$y = (1 + e^{-x})$$

$$\sigma(x) = y^{-1}$$

$$\sigma'(x) = \frac{-1}{y^2} \cdot \frac{dy}{dx} = \frac{-1}{y^2} \cdot (-1 \cdot e^{-x})$$

$$\sigma'(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$$

$$1 - \sigma(x) = \frac{e^{-x}}{(1 + e^{-x})}$$

$$\text{So, } \sigma^-(x) = \sigma(x) \cdot (1 - \sigma(x))$$

$$\textcircled{4} \quad f(x) = \frac{2x+1}{x^2+3}$$

$$\Rightarrow f(x) = (2x+1) \cdot (x^2+3)^{-1}$$

$$f'(x) = 2 \cdot (x^2+3)^{-1} + 2x \cdot (-1 \cdot (x^2+3)^{-2}) \cdot (2x+1)$$

$$= \frac{2}{x^2+3} - \left(\frac{4x^2+2x}{(x^2+3)^2} \right)$$

$$= \frac{2x^2+6 - 4x^2 - 2x}{(x^2+3)^2}$$

$$= \frac{-2x^2 - 2x + 6}{(x^2+3)^2}$$

$$\textcircled{5} \quad f(x) = 4x^5 - 3x^2 + 8$$

$$f'(x) = 20x^4 - 6x$$

$$\textcircled{6} \quad f(x) = \ln(5x^2 - 3x)$$

$$f'(x) = \frac{1}{5x^2 - 3x} \cdot (10x - 3)$$

$$= \frac{10x - 3}{5x^2 - 3x}$$

$$\textcircled{7} \quad f(x) = (x^2 + 1)(3x - 2)$$

$$f'(x) = 2x \cdot (3x - 2) + (x^2 + 1)(3)$$

$$= 6x^2 - 2x + 3x^2 + 3$$

$$= 9x^2 - 2x + 3$$

$$\quad \quad \quad \rightarrow 4x$$

⑧ $f(x) = (4x^3 - 2x)^5$

$$f'(x) = 5 \cdot (4x^3 - 2x) \cdot (12x^2 - 2)$$